

Dose-Specific Beta-Binomial Efficacy Model

with a Dose-Specific IPDE Carryover Parameter

Purpose. This note specifies a dose-specific Bayesian model for a binary efficacy endpoint in regular and IPDE patients. At each dose, the regular-patient efficacy probability receives a Beta prior, while the IPDE probability is linked to it through a dose-specific efficacy carryover parameter. The priors are independent across dose levels; no hierarchical borrowing across doses is assumed.

1. Data structure and notation

Let the ordered dose levels be indexed by $j = 1, \dots, J$. Superscripts (R) and (I) denote regular and IPDE patients, respectively, and the subscript E denotes the efficacy endpoint.

Quantity	Definition
$n_{E,j}^{(R)}, y_{E,j}^{(R)}$	Number of regular patients treated at dose j and number with an efficacy response.
$n_{E,j}^{(I)}, y_{E,j}^{(I)}$	Number of IPDE patients treated at dose j and number with an efficacy response.
$p_{E,j}^{(R)}$	Efficacy probability for a regular patient treated at dose j .
$p_{E,j}^{(I)}$	Efficacy probability for an IPDE patient treated at dose j .
$r_{E,j}$	Dose-specific efficacy carryover parameter for an IPDE patient at dose j , defined for $j \geq 2$.

Conditional on the probabilities, the observed efficacy counts follow

$$Y_{E,j}^{(R)} \mid p_{E,j}^{(R)} \sim \text{Binomial}\left(n_{E,j}^{(R)}, p_{E,j}^{(R)}\right), \quad (1)$$

$$Y_{E,j}^{(I)} \mid p_{E,j}^{(I)} \sim \text{Binomial}\left(n_{E,j}^{(I)}, p_{E,j}^{(I)}\right). \quad (2)$$

The counts are conditionally independent across patient types and dose levels given the model parameters.

2. Dose-specific IPDE carryover model

For doses $j = 2, \dots, J$, define

$$p_{E,j}^{(I)} = p_{E,j}^{(R)} + \left(1 - p_{E,j}^{(R)}\right) r_{E,j} = 1 - \left(1 - p_{E,j}^{(R)}\right) (1 - r_{E,j}). \quad (3)$$

Thus, $r_{E,j}$ is the additional probability of efficacy attributable to IPDE carryover among patients who would not respond to the current dose alone.

- $r_{E,j} = 0$ implies no efficacy carryover and $p_{E,j}^{(I)} = p_{E,j}^{(R)}$.

- $0 < r_{E,j} < 1$ implies $p_{E,j}^{(R)} < p_{E,j}^{(I)} < 1$.
- $r_{E,j} = 1$ implies $p_{E,j}^{(I)} = 1$.

Dose 1 has no preceding lower-dose exposure. Under the current design, IPDE is not permitted at dose 1, so $n_{E,1}^{(I)} = 0$ and no $r_{E,1}$ is estimated. If a software implementation requires an IPDE probability at dose 1, define $p_{E,1}^{(I)} = p_{E,1}^{(R)}$ only as a bookkeeping constraint.

Structural implication. Equation (3) requires $p_{E,j}^{(I)} \geq p_{E,j}^{(R)}$. It is appropriate only when prior exposure is assumed to preserve or increase efficacy. If IPDE patients may have lower efficacy than regular patients at the same dose, the two probabilities should instead be modeled separately or linked with a model that permits effects in both directions.

3. Dose-specific Beta priors

Assign independent priors

$$p_{E,j}^{(R)} \sim \text{Beta}(a_{p,j}, b_{p,j}), \quad j = 1, \dots, J, \quad (4)$$

$$r_{E,j} \sim \text{Beta}(a_{r,j}, b_{r,j}), \quad j = 2, \dots, J. \quad (5)$$

The notation $a_{p,j}, b_{p,j}$ is reserved for the regular-patient efficacy probability, and $a_{r,j}, b_{r,j}$ is reserved for the efficacy carryover parameter. This avoids using the letter R both for “regular patient” and for prior hyperparameters.

For $X \sim \text{Beta}(a, b)$,

$$\mathbb{E}(X) = \frac{a}{a+b}, \quad \text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)},$$

and $a+b$ is commonly interpreted as the prior concentration or prior effective sample size.

Terminology. A binomial likelihood with a Beta prior has a beta-binomial marginal distribution after the probability is integrated out. In the linked IPDE model, the regular-patient component is beta-binomial marginally, but the joint model for regular and IPDE data is not a standard conjugate beta-binomial model because $p_{E,j}^{(I)}$ is a deterministic function of both $p_{E,j}^{(R)}$ and $r_{E,j}$.

4. Likelihood and posterior distribution

Let

$$\mathbf{p}_E^{(R)} = (p_{E,1}^{(R)}, \dots, p_{E,J}^{(R)}), \quad \mathbf{r}_E = (r_{E,2}, \dots, r_{E,J}).$$

Ignoring binomial coefficients and using $n_{E,1}^{(I)} = 0$, the likelihood is

$$\begin{aligned} L(\mathbf{p}_E^{(R)}, \mathbf{r}_E; \mathcal{D}) &\propto \prod_{j=1}^J \left(p_{E,j}^{(R)} \right)^{y_{E,j}^{(R)}} \left(1 - p_{E,j}^{(R)} \right)^{n_{E,j}^{(R)} - y_{E,j}^{(R)}} \\ &\quad \times \prod_{j=2}^J \left[p_{E,j}^{(R)} + (1 - p_{E,j}^{(R)}) r_{E,j} \right]^{y_{E,j}^{(I)}} \left[(1 - p_{E,j}^{(R)}) (1 - r_{E,j}) \right]^{n_{E,j}^{(I)} - y_{E,j}^{(I)}}. \end{aligned} \quad (6)$$

The joint posterior is

$$\begin{aligned} \pi(\mathbf{p}_E^{(R)}, \mathbf{r}_E \mid \mathcal{D}) &\propto L(\mathbf{p}_E^{(R)}, \mathbf{r}_E; \mathcal{D}) \prod_{j=1}^J \left(p_{E,j}^{(R)}\right)^{a_{p,j}-1} \left(1 - p_{E,j}^{(R)}\right)^{b_{p,j}-1} \\ &\quad \times \prod_{j=2}^J \left(r_{E,j}\right)^{a_{r,j}-1} \left(1 - r_{E,j}\right)^{b_{r,j}-1}. \end{aligned} \quad (7)$$

For a single dose $j \geq 2$, the posterior kernel can be written explicitly as

$$\begin{aligned} \pi(p_{E,j}^{(R)}, r_{E,j} \mid \mathcal{D}_j) &\propto \left(p_{E,j}^{(R)}\right)^{y_{E,j}^{(R)} + a_{p,j} - 1} \left(1 - p_{E,j}^{(R)}\right)^{n_{E,j}^{(R)} - y_{E,j}^{(R)} + n_{E,j}^{(I)} - y_{E,j}^{(I)} + b_{p,j} - 1} \\ &\quad \times \left[p_{E,j}^{(R)} + \left(1 - p_{E,j}^{(R)}\right) r_{E,j}\right]^{y_{E,j}^{(I)}} \left(r_{E,j}\right)^{a_{r,j} - 1} \left(1 - r_{E,j}\right)^{n_{E,j}^{(I)} - y_{E,j}^{(I)} + b_{r,j} - 1}. \end{aligned} \quad (8)$$

Because of the term in square brackets, this posterior is not a closed-form Beta distribution. MCMC or numerical integration is therefore required.

5. JAGS specification

The following JAGS model implements the parameterization above. The arrays `a_p`, `b_p`, `a_r`, and `b_r` may vary by dose.

```
model {
  for (j in 1:ndose) {
    pE_regular[j] ~ dbeta(a_p[j], b_p[j])
    yE_regular[j] ~ dbin(pE_regular[j], nE_regular[j])
  }

  # No IPDE carryover is estimated at dose 1.
  pE_ipde[1] <- pE_regular[1]
  yE_ipde[1] ~ dbin(pE_ipde[1], nE_ipde[1])

  for (j in 2:ndose) {
    rE[j] ~ dbeta(a_r[j], b_r[j])
    pE_ipde[j] <- pE_regular[j] +
      (1 - pE_regular[j]) * rE[j]
    yE_ipde[j] ~ dbin(pE_ipde[j], nE_ipde[j])
  }
}
```

Monitor `pE_regular`, `pE_ipde`, and `rE`. If the design prohibits IPDE at dose 1, pass `nE_ipde[1] = 0` and `yE_ipde[1] = 0`.

6. Interpretation and identifiability

Separate roles of the parameters. $p_{E,j}^{(R)}$ represents efficacy from the current dose for a regular patient. The parameter $r_{E,j}$ represents the additional IPDE carryover contribution. The induced IPDE probability $p_{E,j}^{(I)}$ is a derived quantity, not an independently assigned parameter.

Data required to estimate carryover. At dose j , regular and IPDE observations are both needed to distinguish $p_{E,j}^{(R)}$ from $r_{E,j}$. In particular:

- If $n_{E,j}^{(I)} = 0$, the posterior distribution of $r_{E,j}$ remains its prior.
- If $n_{E,j}^{(R)} = 0$, $p_{E,j}^{(R)}$ and $r_{E,j}$ are strongly confounded because the IPDE data identify primarily the combination $p_{E,j}^{(I)}$.
- With small sample sizes in either group, inference for $r_{E,j}$ will be prior-sensitive and credible intervals may be wide.

No borrowing across doses. The priors above are dose-specific and mutually independent. Information at dose k does not update $p_{E,j}^{(R)}$ or $r_{E,j}$ for $k \neq j$. A hierarchical prior would be a separate extension and is not part of this model.

7. Relation to an α -based data-generating mechanism

For any valid pair satisfying $p_{E,j}^{(R)} \leq p_{E,j}^{(I)} \leq 1$ and $p_{E,j}^{(R)} < 1$, the equivalent carryover parameter is

$$r_{E,j} = \frac{p_{E,j}^{(I)} - p_{E,j}^{(R)}}{1 - p_{E,j}^{(R)}}. \quad (9)$$

If simulation data are generated by

$$p_{E,j}^{(I)} = p_{E,j}^{(R)} + \alpha p_{E,j-1}^{(R)},$$

then, provided $p_{E,j}^{(R)} + \alpha p_{E,j-1}^{(R)} \leq 1$,

$$r_{E,j} = \frac{\alpha p_{E,j-1}^{(R)}}{1 - p_{E,j}^{(R)}}. \quad (10)$$

If the generated IPDE probability is truncated at 1, the equivalent value is $r_{E,j} = 1$ whenever the truncation is active.

8. Posterior summaries and use in dose selection

For each dose, report posterior summaries of

$$p_{E,j}^{(R)}, \quad p_{E,j}^{(I)}, \quad r_{E,j},$$

including posterior means or medians and 95% credible intervals. A clinically interpretable efficacy criterion is

$$\Pr(p_{E,j}^{(R)} > \phi_E \mid \mathcal{D}),$$

where ϕ_E is the prespecified efficacy threshold.

For final dose evaluation or utility calculation, $p_{E,j}^{(R)}$ is generally the relevant efficacy probability because it represents the expected response for a new patient receiving dose j without prior IPDE exposure. The derived probability $p_{E,j}^{(I)}$ is used to model outcomes among IPDE patients and to estimate the carryover effect.

Model summary. The model separates the regular-patient efficacy probability $p_{E,j}^{(R)}$ from the IPDE carryover parameter $r_{E,j}$ and defines $p_{E,j}^{(I)} = p_{E,j}^{(R)} + (1 - p_{E,j}^{(R)})r_{E,j}$. The parameterization is bounded and interpretable, but it imposes the one-sided assumption $p_{E,j}^{(I)} \geq p_{E,j}^{(R)}$.